ChatGLM-FGIDs-TCM: A Knowledge-Integrated Large Language Model for Clinical Decision Support in Traditional Chinese Medicine

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***Abstract*—Functional Gastrointestinal Disorders (FGIDs) represent some of the most prevalent gastrointestinal conditions worldwide. While first-line pharmacological treatments exist, a significant number of patients—particularly those with stress-related or psychosomatic symptoms—achieve only limited relief. Traditional Chinese Medicine (TCM) provides a holistic therapeutic alternative with distinct characteristics. This study presents ChatGLM-FGIDs-TCM, a domain-adapted large language model based on the open-source ChatGLM-6B framework. By incorporating Retrieval-Augmented Generation (RAG), a specialized TCM knowledge base, a structured knowledge graph, and terminology normalization through word networks, the model enhances reasoning capabilities and clinical applicability for TCM Clinical Decision Support Systems (TCM-CDSS). Comprehensive evaluations involving both objective tasks with varying difficulty and subjective assessments demonstrate that ChatGLM-FGIDs-TCM substantially outperforms baseline LLMs and exceeds the performance of junior physicians in specific reasoning tasks. The model shows considerable promise for deployment in TCM education, public health initiatives, and initial clinical triage.**

Keywords—Traditional Chinese Medicine, Functional Gastrointestinal Disorders, Large Language Model, Clinical Decision Support, Knowledge Graph, Retrieval-Augmented Generation

Introduction

Functional gastrointestinal disorders (FGIDs) comprise a group of conditions characterized by diverse functional alterations in the absence of identifiable organic disease that would explain the clinical symptoms. These disorders represent a common and frequently encountered category within digestive system diseases. A 2021 global epidemiological survey indicated that FGID prevalence generally exceeds 40% in Europe and the United States, while ranging from 30.0% to 39.9% in China [1]. Pharmacological therapy serves as the primary treatment approach for FGIDs. Nevertheless, previous research demonstrates that first-line drug treatments fail to adequately improve symptoms in many patients, particularly those with neuroticism, stress-related symptoms, and physical disorders [2]. Traditional Chinese Medicine (TCM), which employs a comprehensive methodology grounded in TCM theory and clinical symptom analysis, has demonstrated potential advantages in managing FGIDs [3]. This integrated approach can yield superior therapeutic outcomes.

The Traditional Chinese Medicine Clinical Decision Support System (TCMCDSS) represents a crucial component of TCM intelligence. It assists physicians by integrating patient data, medical knowledge, and case-specific guidelines to enhance clinical reasoning while improving efficiency and competence. Existing TCMCDSS implementations span multiple aspects of the TCM diagnostic and treatment process, including tongue diagnostic instruments and facial diagnostic instruments based on color space models [4], intelligent TCM inquiry systems utilizing natural language processing [5], and TCM pulse diagnosis robots employing array sensors with pulse pattern recognition and analysis [6]. However, current TCM clinical decision support systems rely predominantly on rule-based algorithms or conventional machine learning techniques. These systems encounter limitations due to constrained training data availability and inconsistent data quality. Such constraints hinder the systems' capacity for evolution and improvement in diagnostic and treatment accuracy. The scarcity of comprehensive datasets coupled with data quality variability can introduce biases into the decision-support process, presenting significant challenges for developing reliable TCM clinical decision support systems. Addressing these issues remains essential for advancing TCMCDSS development and integrating TCM practices with modern healthcare technologies.

In recent years, large language models (LLMs) have achieved remarkable performance across diverse natural language processing tasks [7], while demonstrating human-like knowledge inference capabilities [8]. Autoregressive large language models such as InstructGPT [9], ChatGPT [10], and GPT-4 [11] employ pre-training and fine-tuning techniques to understand and follow human instructions, enabling them to comprehend and respond accurately to complex questions.

These developments have extended into the medical domain. PMC-LLaMA introduced a pre-trained language model based on biomedical literature [13], while MedPaLM established the MultiMedQA benchmark for medical question answering that covers medical examinations, research, and consumer health inquiries [14]. ChatDoctor fine-tuned the LLaMA model using medical domain knowledge to create a specialized medical chat model [15]. Hua Tuo (Materia Medica) developed a LLaMA fine-tuning model based on Chinese medical knowledge, primarily referencing the

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Chinese medical knowledge graph CMeKG [16,17]. DoctorGLM constitutes a Chinese medical consultation model based on ChatGLM-6B that enables ChatGPT to generate responses incorporating medical guideline information [18]. TCMChat represents a generative LLM pre-trained and fine-tuned on extensive curated TCM texts and Chinese question-answer datasets [19]. The Lingdan-TCPM-Chat model utilizes chain-of-thought reasoning for symptom analysis and TCPM recommendations, while the Lingdan Prescription Recommendation Model (Lingdan-PR) generates herbal prescriptions based on electronic medical records [20].

The emergence of large language models presents a significant opportunity for TCMCDSS development, though representative studies applying LLMs to TCMCDSS remain scarce. Within this context, this study investigates an intelligent diagnosis and treatment system for FGIDs to explore LLM applications in TCMCDSS, with the aim of providing inspiration for future related research.

Data and methods

I. Technical Framework Overview

We employ Retrieval-Augmented Generation (RAG) as the core framework to construct a robust TCM-CDSS, enhancing the factual accuracy and reasoning ability of large language models through the integration of external domain knowledge. The proposed system, ChatGLM-FGIDs-TCM, incorporates a multi-source knowledge base, a domain-specific knowledge graph, and a terminology-normalization word network, establishing a dual-layer enhancement mechanism.

As illustrated in Figure 1, the RAG pipeline initiates with user queries and performs document retrieval from a curated TCM corpus. Retrieved documents are processed by a fine-tuned ChatGLM-6B model, guided by prompt templates optimized for TCM question-answering scenarios.

To further improve answer quality, model outputs are processed through a word network that aligns synonyms, hypernyms, and TCM terminologies to a standardized vocabulary. These normalized terms subsequently activate corresponding nodes in the TCM knowledge graph, introducing structured domain reasoning into the response generation cycle.

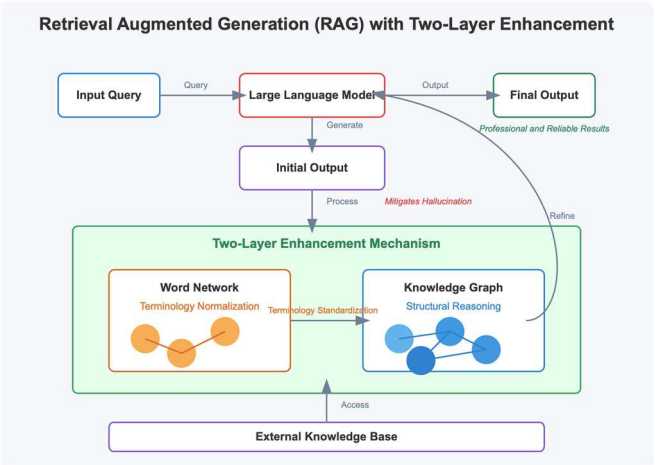


Figure 1 Retrieval Augmented Generation (RAG) with Two-Layer Enhancement

II. Knowledge Graph Construction

Based on the Knowledge Graph (KG) constructed by our previous research[21], the case data of real-world patients in the Department of Spleen and Stomach Diseases of Guang 'anmen Hospital of China Academy of Chinese Medical Sciences from 2014 to 2023 were used as the new data source. The entities and relationships were extracted, and 10326 entity-relationship-entity triples were constructed. The top-down construction method was used to expand the TCM knowledge map of FGIDs, which laid a knowledge foundation for subsequent research and application.

In the process of designing the structure of knowledge graph, we refer to the knowledge inference process of TCM doctors in real clinical situations. We divide this process into three types of sub-inference processes: main inference, auxiliary inference and supplementary inference. The main inference refers to the traditional inference process of symptom - disease - pattern – treatment method - prescription - herb, that is, the process of abstracting concrete symptoms into traditional Chinese medicine theory and then restoring concrete herbs. In the main process, the inference of some stages needs to combine the information of other stages. We call the inference process that combines the information of other stages as auxiliary inference. After the conclusion of the main inference process, Chinese medicine practitioners often supplement some herbs for the special symptoms of patients, we call this process supplementary inference. Through this knowledge graph, we can learn all the basic information needed by TCM doctors when making decisions, and then restore the inference process of TCM doctors, and enhance the inference ability of large language models. The graphical representation of the knowledge graph structure is as follows.

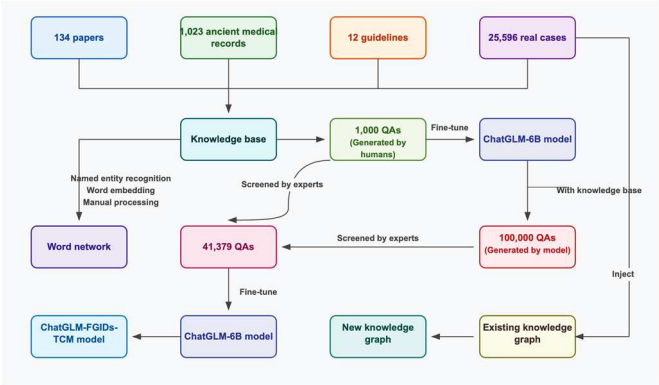


Figure 2 Construction process of inference model

The inference process of the model is shown as follows (Figure 3) :

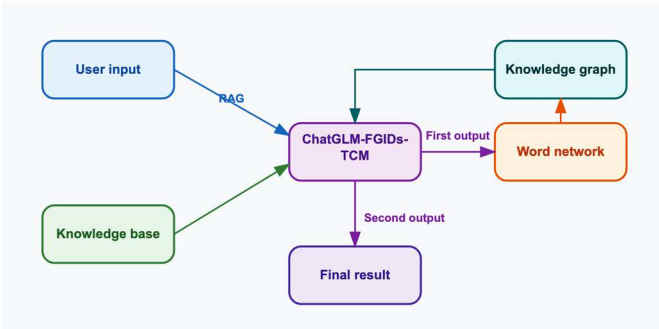


Figure 3 Model inference process

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III. Knowledge Base and Data Augmentation

ChatGLM-6B[22] is an open source conversational language model that supports Chinese and English bilingual Q&A and is optimized for Chinese. The Model is based on the General Language Model (GLM) architecture and has 6.2 billion parameters. Combined with model quantization technology, it can be deployed locally on consumer graphics cards and has excellent inference performance. Therefore, combined with the actual situation of the laboratory, ChatGLM-6B was selected as the base model in this study.

We compiled a comprehensive TCM knowledge base by retrieving resources from the Science Citation Index, Engineering Index, CSCD, and the Ancient and Modern Medical Cases Cloud Platform [23], which includes:

- 25,596 modern clinical records
- 1,023 ancient medical cases
- 134 journal articles
- 12 clinical guidelines (both TCM and Western)

To facilitate effective supervised fine-tuning (SFT), a dataset of high-quality, task-specific question-answer pairs was essential. We initiated this process by having TCM experts manually construct a seed set of 1,000 diverse and challenging QA pairs covering syndromes, prescriptions, herbs, theoretical principles, and diagnostic rules. This seed set was used to perform an initial fine-tuning of the base ChatGLM-6B model. The fine-tuned model was then leveraged to generate a large-scale set of 100,000 synthetic QA pairs, a process that amplifies the diversity and scope of the training data. This synthetic dataset underwent a rigorous manual screening and curation process by our TCM specialists to filter out errors, inconsistencies, and low-quality samples. The outcome was a refined, high-quality SFT dataset containing 41,379 QA pairs. The fine-tuning process was executed using the parameter-efficient P-Tuning v2 method [24], which maintains strong performance while significantly reducing computational cost compared to full-parameter fine-tuning.

IV. Prompt Design and Search-Enhanced Generation

The LangChain framework was used to integrate the LLM with the TCM knowledge base and to design context-aware prompts. The standard prompt template is as follows:

> Known information: {search_results}
> You are a Chinese medicine functional gastrointestinal disease intelligent question and answer assistant, please answer the user's questions according to the above known information, concise and professional, the answer must quote the known information, do not appear in the answer of the specific patient information. If you can't get an answer from it, please say "can't answer the question", do not allow to add fabrications in the answer, please use Chinese.
> Question: {question}

This method confines responses to retrieved evidence, reducing hallucinations.

V. Word Network Integration and KG Feedback Loop

To address the challenge of lexical variability in TCM terminology (e.g., "腹胀" and "肚子胀" both meaning abdominal distension), we constructed a dedicated word network. This network acts as a normalization layer, mapping diverse surface forms of terms found in user queries and model outputs to a standardized set of clinical terminologies. The mapping leverages semantic similarity computed using pre-trained word embeddings (e.g., Word2Vec, BERT) and was subsequently validated and refined through manual verification by TCM experts to ensure clinical accuracy.

The normalized terms output by this network are then used as queries to the TCM knowledge graph. The relevant sub-graphs—containing interconnected nodes and relationships—are retrieved. These structured semantic paths (e.g., the path from a specific symptom to its associated pattern and recommended formula) are formatted into a structured prompt and fed back into the language model in a subsequent step. This creates a feedback loop where the initial, neurally generated response is enriched and constrained by expert-defined, symbolic knowledge paths from the KG, augmenting the final response with a layer of explicit, structured clinical reasoning.

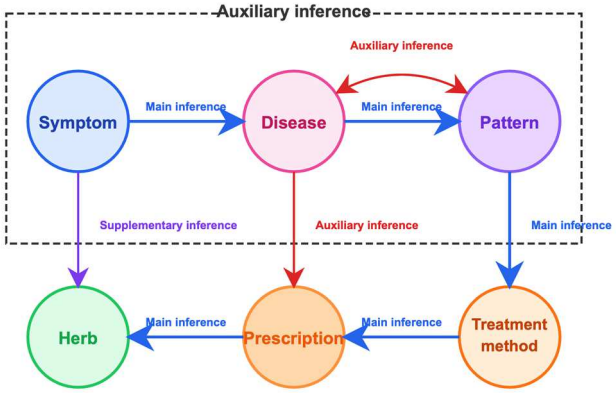


Figure 4 Knowledge graph structure

VI. Evaluation Strategy

A rigorous and multi-faceted evaluation was designed to assess the performance of ChatGLM-FGIDs-TCM from various angles. The system was evaluated against the following benchmarks:

1） Baseline LLMs: Three powerful general-purpose and domain-specific LLMs: GPT-3.5 [25] (via API), ERNIE-3.5 [26] (a leading Chinese LLM), and the recently released Qwen-2.0 [27].

2） Human Experts: Five licensed TCM physicians with varying levels of clinical experience (categorized as junior and senior for analysis).

3） Ablation Configurations: To isolate the contribution of each key component, we evaluated two ablated versions of our system:

- ChatGLM-FGIDs-TCM-RKG: The system operating without the knowledge graph feedback loop.

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- ChatGLM-FGIDs-TCM-RKB: The system operating without the initial retrieval from the external knowledge base.

Performance was assessed through two primary modalities:

1） Objective Evaluation: A benchmark of 60 multiple-choice questions was developed. This set included 30 Simple Questions (SQs) designed to test direct recall of clinical or textbook knowledge, and 30 Difficult Questions (DQs) that required multi-step logical reasoning, synthesis of multiple concepts, or nuanced pattern differentiation. The questions were meticulously crafted and validated by three senior TCM specialists specializing in spleen and stomach diseases. The key metric was accuracy, defined as the proportion of correctly answered items per category.

2） Subjective Evaluation: Thirty open-ended, free-form questions were used, simulating complex clinical queries. The same panel of TCM experts scored the responses from all models and the junior physicians in a single-blinded manner. They assigned scores from 0 to 10 based on a predefined rubric focusing on factual accuracy, clinical relevance and applicability, coherence and fluency of explanation, and adherence to TCM principles. The mean expert score served as the primary metric for this evaluation.

Results

I. Data Summary

The finalized TCM knowledge graph for FGIDs contained 10,326 structured entities interconnected by 112,668 semantic relationships, forming a dense network of domain knowledge. The literature-derived knowledge base comprised a substantial corpus: 25,596 modern clinical cases, 1,023 historical case records from classical texts, 134 academic journal articles, and 12 documents of expert consensus or clinical guidelines. The dataset for fine-tuning and evaluation consisted of 41,379 high-quality question-answer pairs after rigorous manual screening.

II. Objective Evaluation

The performance on the multiple-choice benchmark is summarized in Figure 5. ChatGLM-FGIDs-TCM demonstrated superior performance, significantly outperforming all baseline LLMs and the ablated versions. It achieved a notable accuracy of 86.7% on Simple Questions and 66.7% on Difficult Questions. This performance not only substantially exceeded that of its base model, ChatGLM-6B, but also approached the average performance level of the human physicians. Interestingly, the model's performance on Difficult Questions surpassed its performance on Simple Questions, suggesting that its integrated knowledge structures are particularly effective for complex reasoning tasks. The ablation studies confirmed the value of each component: both the RKG and RKB variants showed a marked drop in accuracy, with the removal of the knowledge base (RKB) having a slightly more pronounced negative effect.Sample questions are provided in Table 1.

TABLE I. Objective Sample Questions

Question type	Question content
Simple question	The key pathogenesis of stomachache is: A. Qi Deficiency B. Qi Reversal C. Qi Sink D. Qi Stagnation
	Symptoms include decreased appetite, emaciation, abdominal distension, and loose stools; the common facial color is: A. Sallow and yellow B. Pale yellow C. Bright yellow D. Dull yellow
Difficult question	Patients with chronic diarrhea can not be cured, abdominal pain is warm, poor appetite, waist acid and cold limbs, tongue is pale and white, pulse sinking late, the appropriate treatment: A. Lizhong Pill B. Sishen Pill C. You GUI Wan D. Human nourishing dirty soup
	Chest, diaphragm and stomach duct discomfort, cold aggravation, heat is reduced, mouth is not thirsty, tongue is pale and white, pulse is slow. The best choice: A. Lilac powder B. Lizhong Pill C. Orange peel bamboo Ru soup D. Five grinding decoction

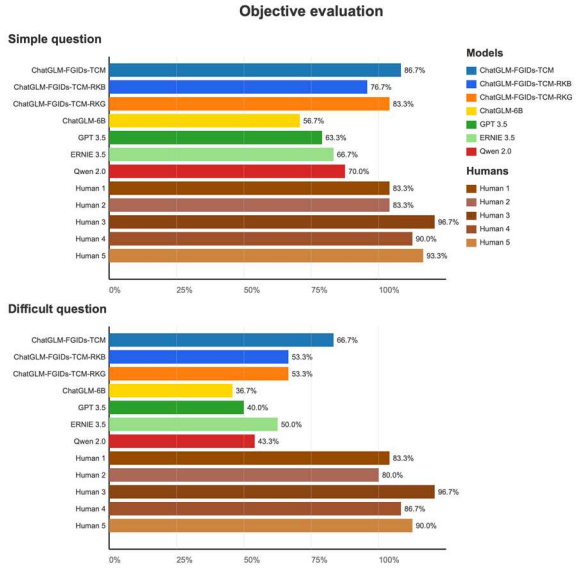


Figure 5 compares the accuracy across all models and physicians.

III. Subjective Evaluation

In the subjective evaluation part, 30 subjective questions with open answers were designed by three experts in spleen and stomach diseases of Chinese medicine, and the three experts scored different answers single-blind, with a full score of 10. The average scores of each model and five human practitioners were calculated as the subjective evaluation standard to verify the effect of the system. Sample questions are as follows (Table 2) .

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TABLE II. SUBJECTIVE SAMPLE QUESTIONS

Discuss "Discomfort in the stomach leads to restless sleep. "
Discuss the "Nine Methods for Treating Diarrhea."

The results of the expert-blind subjective assessment are illustrated in Figure 6. ChatGLM-FGIDs-TCM achieved the highest mean expert score among all the computational models. It exhibited a substantial 33.14% improvement in the mean score over the base ChatGLM-6B model. The qualitative feedback from experts indicated that its responses were not only more accurate but also demonstrated superior coherence, better alignment with established TCM clinical logic, and a stronger ability to provide contextually relevant and nuanced explanations. The ablated models again performed worse, reinforcing the conclusion that both the retrieved textual evidence and the structured KG reasoning are vital for generating high-quality, trustworthy clinical text.

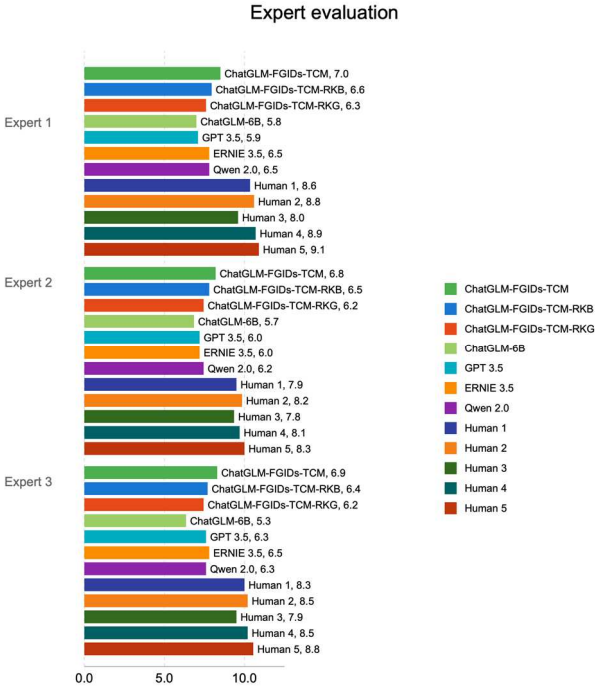


Figure 6 illustrates the expert score comparisons across models and physicians.

The ChatGLM-FGIDs-TCM model exhibited a 33.14% improvement in mean expert score over the base model, demonstrating its enhanced ability to generate coherent, clinically relevant, and context-aware narratives.

IV. DISCUSSION

A. Performance Comparison Across Models

The ChatGLM-FGIDs-TCM model demonstrated consistent and statistically significant superiority over all baseline LLMs, including the robust GPT-3.5 and specialized models like DoctorGLM, in both quantitative and qualitative evaluations. Its notable advantage in handling Difficult Questions underscores a critical point: the integration of structured knowledge via the KG and the grounding of responses via RAG effectively augments the model's capacity for multi-step logical reasoning and complex pattern matching, which are hallmarks of TCM diagnostics. This goes beyond simple information retrieval; it facilitates a form of symbolic-neural reasoning that is essential for navigating the intricate relationships within TCM theory. Compared to general-purpose LLMs, our model's performance highlights the indispensable value of deep, domain-specific adaptation. These models, while powerful in general knowledge, lack the fine-tuned understanding of TCM concepts, the ability to differentiate subtly different patterns, or the knowledge of precise herb-formula relationships, which are crucial for clinical relevance.

B. Human vs. AI Performance in Clinical Contexts

The objective evaluation reveals a fascinating trend: state-of-the-art LLMs are closing the gap with human practitioners on tasks requiring factual knowledge recall (Simple Questions). However, ChatGLM-FGIDs-TCM's ability to surpass the average performance of *junior physicians* on Difficult Questions marks a significant milestone. It suggests that such AI systems can potentially serve as powerful cognitive aids for less experienced practitioners, helping to bridge the experience gap by providing expert-level knowledge inference on demand. Nevertheless, it is crucial to maintain a balanced perspective. The senior TCM experts maintained a clear, albeit narrowed, advantage, particularly in the subjective evaluation where nuance, clinical intuition, and the ability to handle ambiguous or contradictory information come into play. This delineates the current boundaries of AI in medicine: LLMs are exceptionally powerful tools for knowledge retrieval, reasoning support, and handling standardized protocols, but they are not yet replacements for the comprehensive clinical judgment, intuition, and ethical responsibility of an experienced human expert, especially in high-stakes, complex, or atypical cases.

C. Impact of Knowledge Integration: Ablation Results

The ablation studies provide clear, empirical evidence for the architectural choices made in this work. The performance degradation observed in both the RKG and RKB configurations confirms that each component—the unstructured knowledge base and the structured knowledge graph—provides unique and complementary value. The knowledge base serves as a vast reservoir of contextual, descriptive, and case-based information, essential for grounding responses in factual descriptions and clinical evidence. The knowledge graph, on the other hand, provides the "skeleton" of clinical reasoning—the explicit, logical pathways that connect symptoms to treatments. The slightly greater impact of removing the knowledge base suggests that broad textual knowledge is the foundational layer, while the KG provides a critical refinement layer for logical soundness. Together, they create a synergistic effect that is greater than the sum of their parts.

D. Application Potential in Production Environments

Given its demonstrated accuracy, fluency, and domain alignment, ChatGLM-FGIDs-TCM is suitable for deployment in production environments such as public health education, initial symptom screening and triage for functional gastrointestinal disorders (FGIDs), and TCM training or question-answering platforms.

Caution is advised, however, for real-time clinical decision support—especially in cases requiring comprehensive syndrome differentiation, comorbidity management, or treatment personalization. Physician oversight remains strongly recommended during model application.

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In summary, this model represents a notable step toward combining LLMs with domain-specific symbolic reasoning, establishing a foundation for future clinical AI systems in integrative medicine.

V. CONCLUSION

This study presents ChatGLM-FGIDs-TCM, a domain-adapted large language model that synergistically integrates neural language generation with symbolic knowledge structures to advance the state of Traditional Chinese Medicine Clinical Decision Support Systems. By building upon the open-source ChatGLM-6B architecture and enriching it with a multi-source knowledge base, a semantically modeled knowledge graph, and a terminology normalization system within a Retrieval-Augmented Generation framework, the model achieves a significant enhancement in factual accuracy, reasoning depth, and clinical relevance for Functional Gastrointestinal Disorders.

Our comprehensive evaluation, involving both objective metrics and blinded expert subjective assessment, demonstrates that this integrated approach yields substantial performance gains. The model not only outperforms strong baseline LLMs but also surpasses the capabilities of junior TCM physicians in specific, reasoning-intensive tasks. The ablation studies quantitatively validate the contribution of each knowledge component, highlighting the importance of both unstructured textual evidence and structured semantic relationships.

This work provides compelling evidence that hybrid AI systems, which marry the pattern recognition power of large language models with the precision and logic of structured, domain-specific knowledge, represent a highly promising direction for developing intelligent diagnostic and decision-support tools in complex fields like TCM. The model demonstrates immediate practical applicability in educational, public health, and supportive clinical roles.

Despite these successes, several limitations warrant acknowledgment and guide future work. The model's performance, while impressive, does not yet match the nuanced expertise of senior clinicians, particularly in handling complex multi-morbidity cases or integrating information from physical examinations. The reliance on manually curated QA pairs for fine-tuning, though ensuring quality, is resource-intensive and may limit scale. The model's generalizability across the full spectrum of TCM syndromes beyond FGIDs remains to be validated. Furthermore, the current system operates in a standalone manner and is not integrated with live Electronic Health Record (EHR) systems, limiting its utility in real-time clinical workflows.

Future research and development will focus on:

1. **Scalability:** Generalizing the framework to construct TCM-CDSS for other high-prevalence disease domains, such as cardiovascular or respiratory conditions within TCM purview.
2. **Multilingual Expansion:** Incorporating support for multiple languages to facilitate cross-cultural exchange and the global adoption of TCM intelligence.
3. **Real-Time Clinical Integration:** Developing secure interfaces and pipelines to connect the model with hospital EHR systems, enabling it to provide context-aware suggestions based on real patient data.
4. **Advanced Alignment Techniques:** Exploring Reinforcement Learning from Human Feedback (RLHF) or Direct Preference Optimization (DPO) to further refine the model's outputs, ensuring they are not only accurate but also maximally helpful, safe, and aligned with expert clinical preferences and decision-making processes.

In summary, this work proposes and validates a novel, hybrid architecture that effectively merges the capabilities of large language models with the rich, structured knowledge of Traditional Chinese Medicine. ChatGLM-FGIDs-TCM stands as a proof-of-concept and a reference design for future AI systems that seek to harmonize the depth of traditional medical wisdom with the power of contemporary computational intelligence, paving the way for more intelligent, accessible, and effective TCM healthcare solutions.

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